
ELEC6095 Smart Grid

Challenges in Modern Grid Operation

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Contents adopted from: Prof. J. Zhong

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Fundamental of Power Grid Operation and Control

Power System Operation



- Schedule and control outputs of generators.
- Maintain the balance of generation and load.
- Minimize production costs.
- Ensure the transmission system is operating correctly.



Load changes



However, electricity load changes and it is hard to accurately forecast load in advance.



Power generations need to be adjusted in real time to balance changing load.

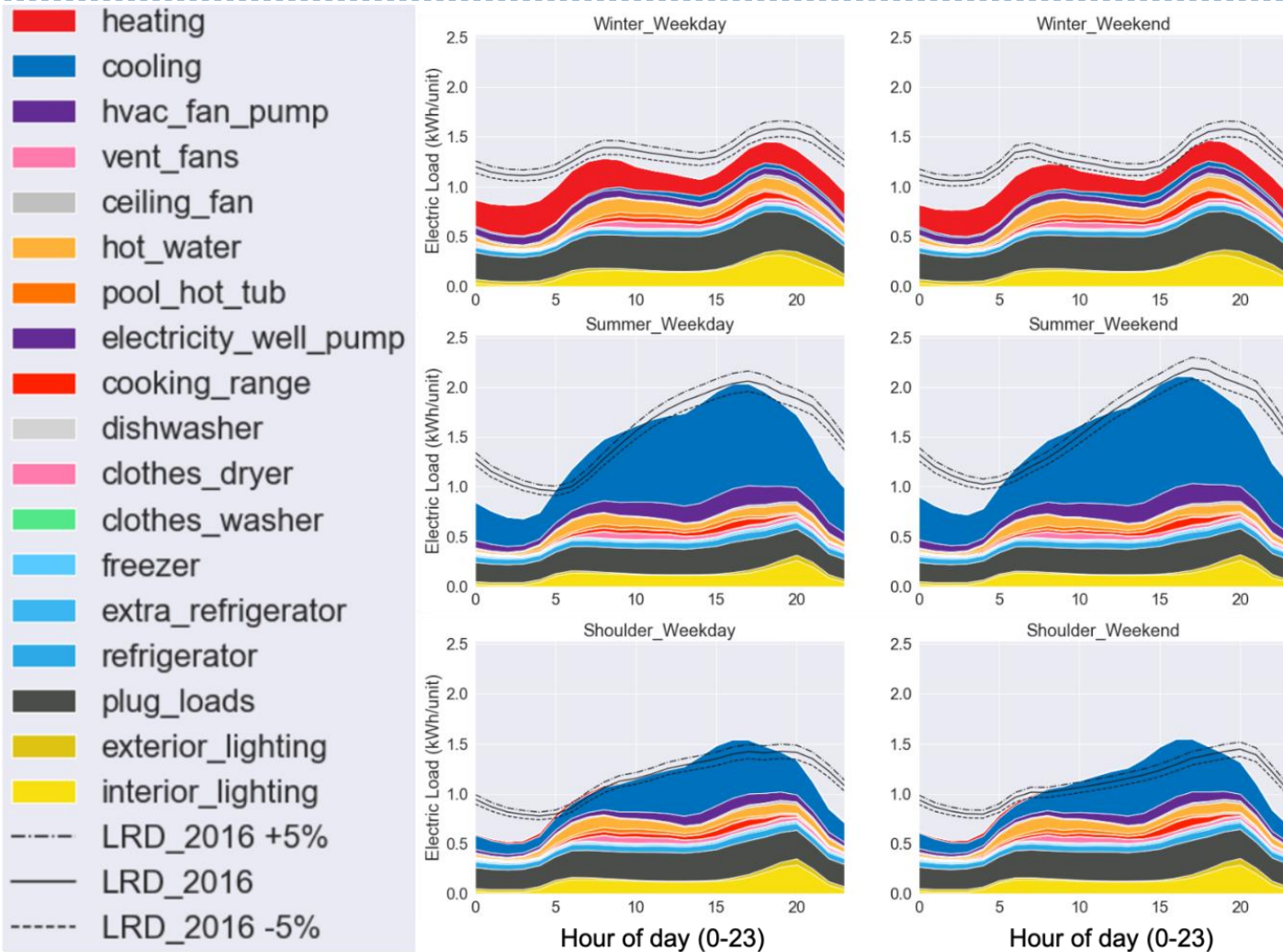


Load change cycles (daily, weekly, seasonally, annually)

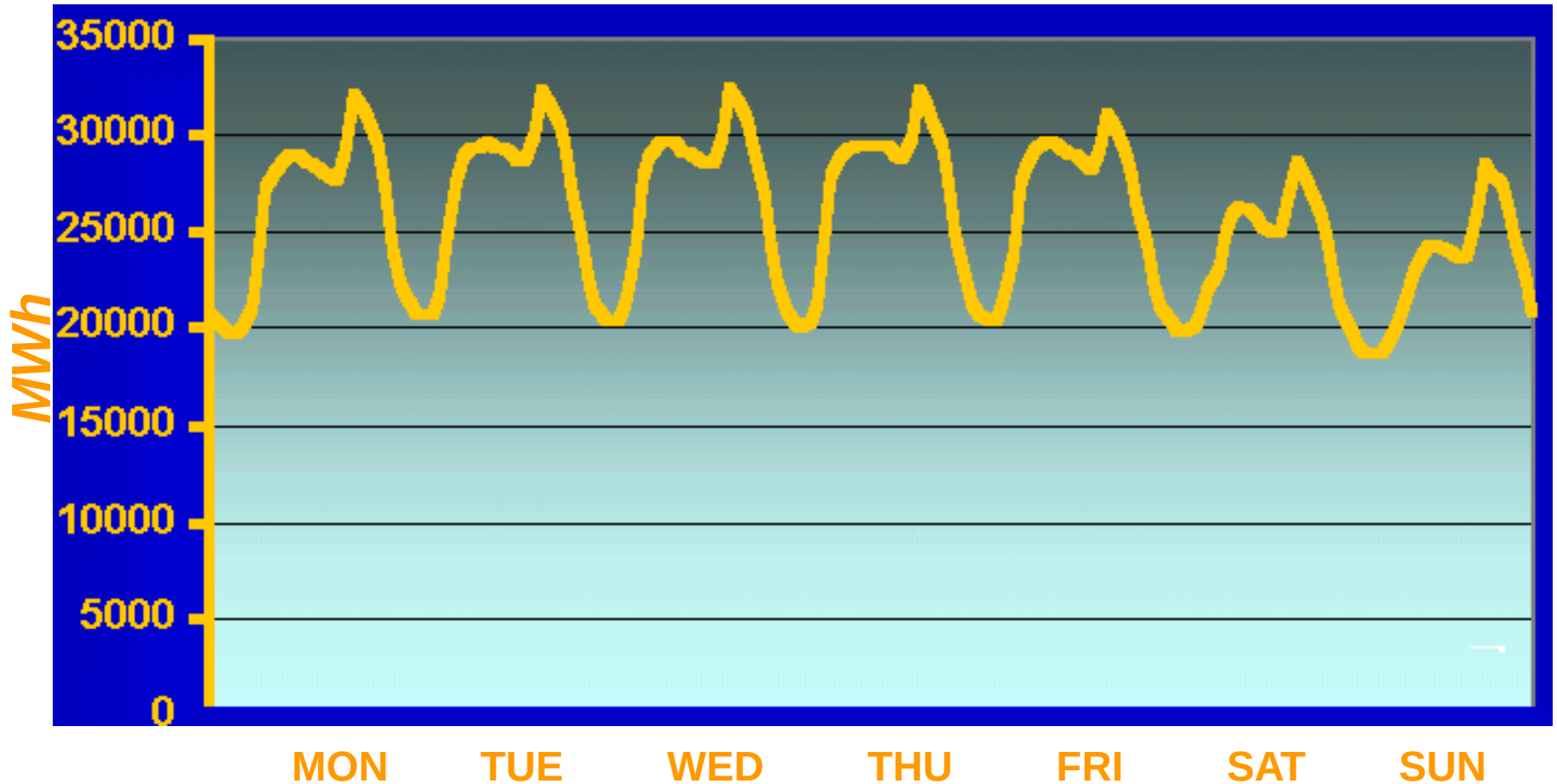


Load profiles

Twenty-four hour load profile

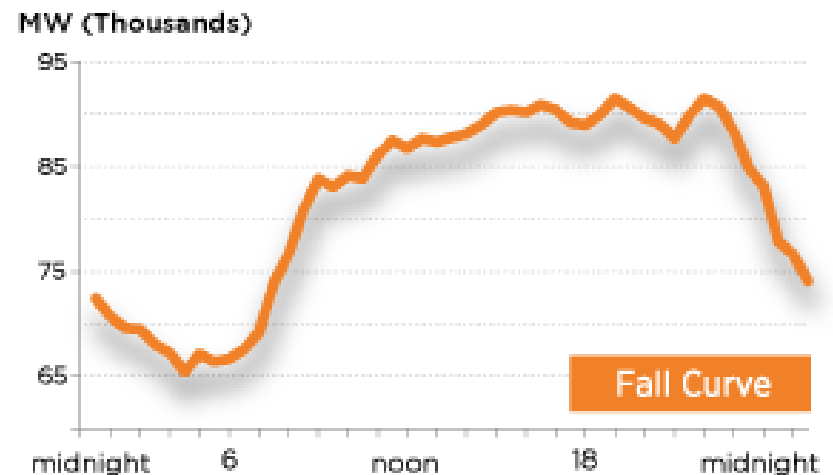
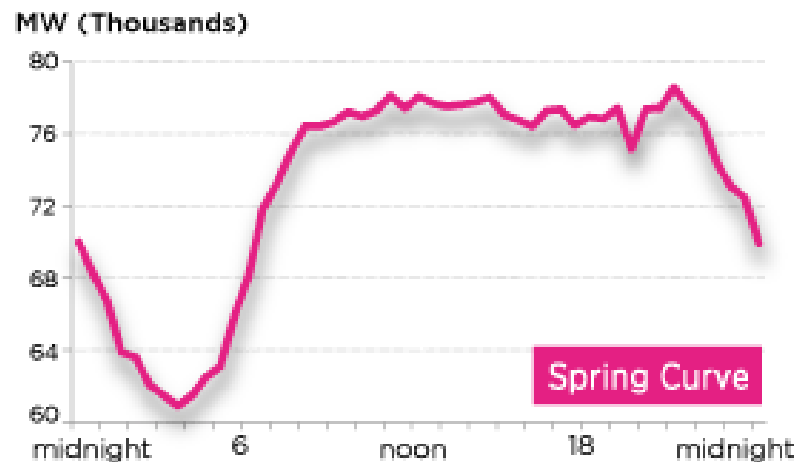
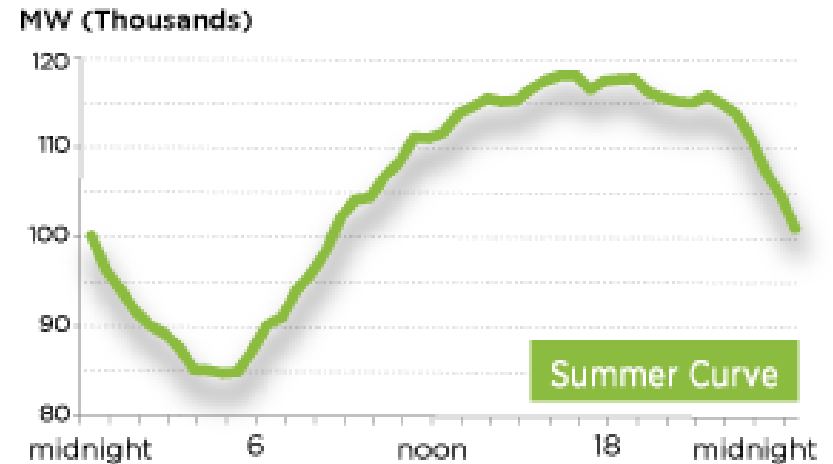
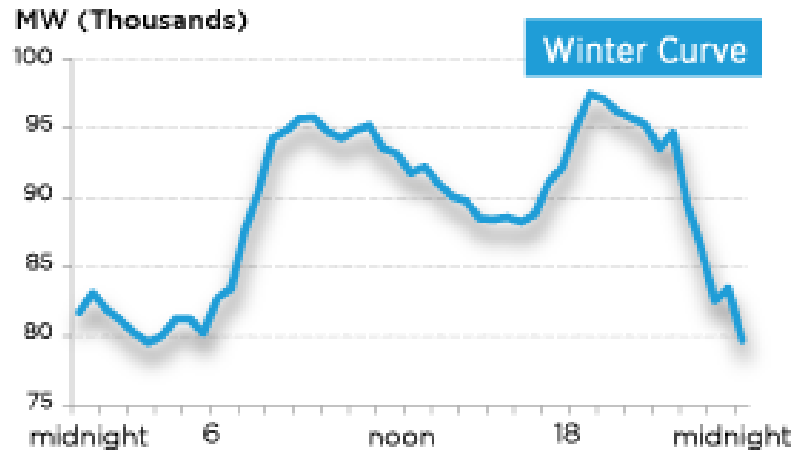


Weekly load cycle



Source: California ISO data

Load patterns for different seasons



Power System Operation

- ▶ Power system operation requires load studies, fault calculations, protection design, and studies of system stability.
- ▶ **An important problem** in system operation is: determining how the total generation required at any time can be distributed among various power plants and within the units of each plant.
- ▶ Applications of computers made great contributions in power system operation and planning.

Power System Planning and Operation



Due to load change cycles, power generations of generators are scheduled in different time intervals.



Long term planning

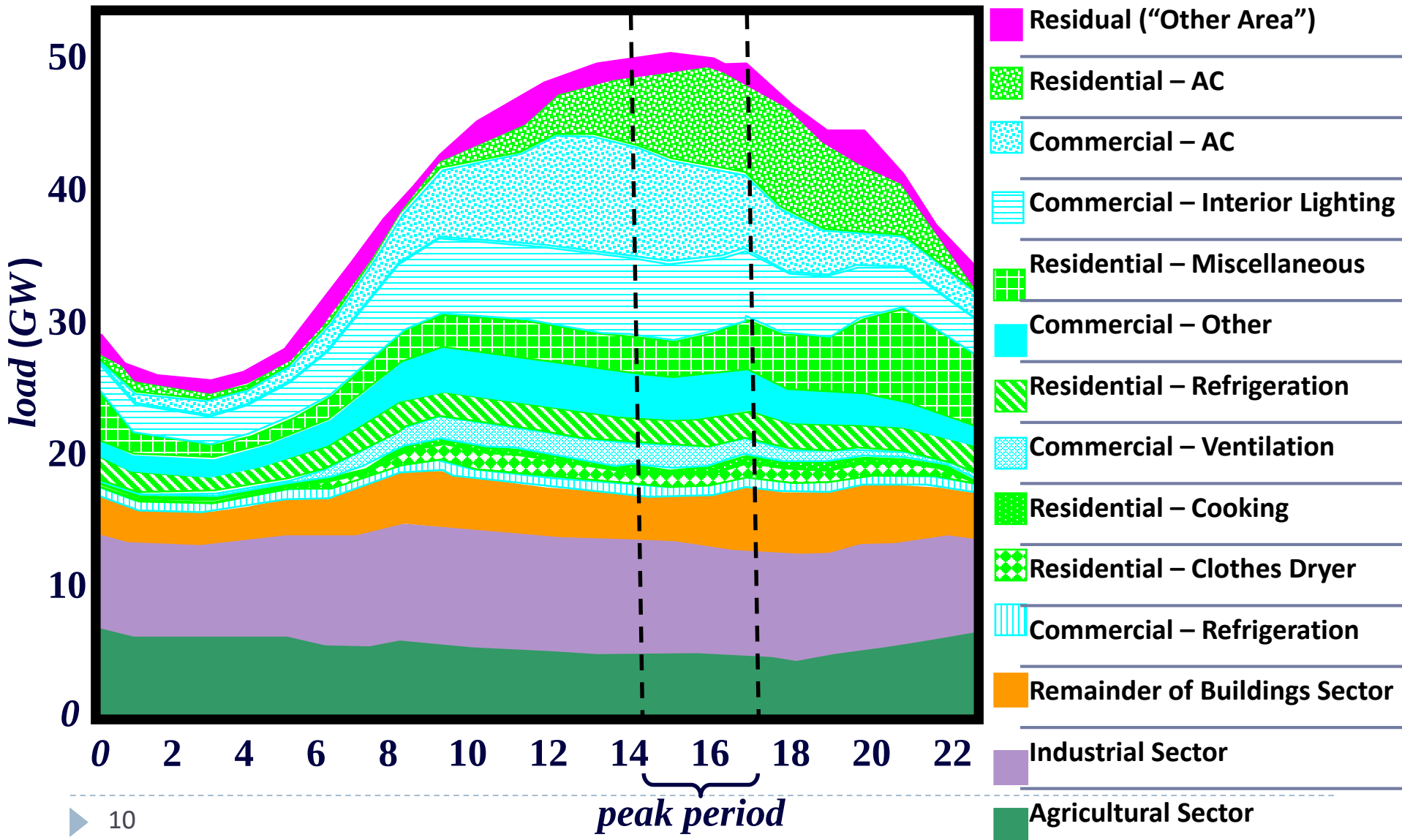


Short-term operation and dispatch

Pre-dispatch
Dispatch
Instantaneous dispatch

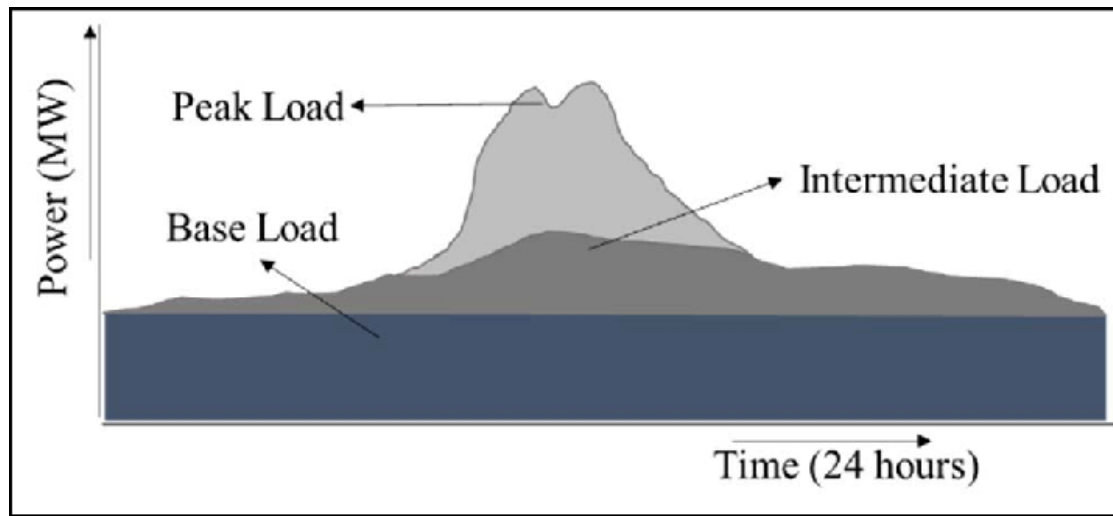


Automation generation control and load following (frequency control)

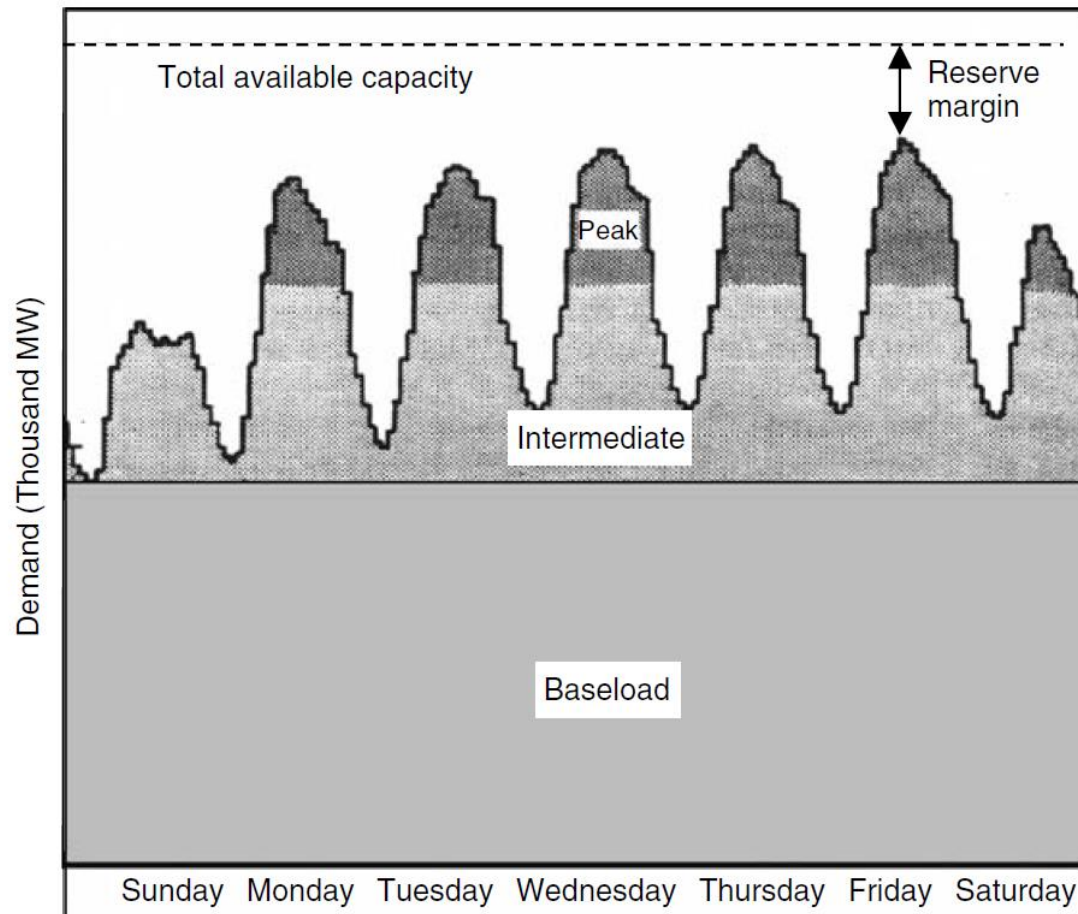


Base load, intermediate and peaking power plants

- ▶ The demand for electricity vary from day to night.
- ▶ Vary from day to day in a week (weekly pattern).
- ▶ Vary from season to season.
- ▶ Total load → base load, intermediate load and peak load



Weekly load fluctuations



Example of weekly load fluctuations and roughly how power plants can be categorized as baseload, intermediate, or peaking plants.

Base load, intermediate and peaking power plants

- ▶ During the peak demand, most of the power plants will be operating, which in the valleys, many are likely to be shut off.
- ▶ Some power plants, such as, **large coal-fired plants and hydro stations**, are expensive to build but relatively cheap to operate. So, they should be run continuously as **base-load power plants**. (nuclear power plant as well)
- ▶ **Simple-cycle gas turbines** are inexpensive to build but expensive to operate. They are most appropriately used as **peaking power plants**. Turn on only during periods of highest demand.

Base load, intermediate and peaking power plants (cont'd)

- ▶ Other plants have characteristics that are in between, serve as *intermediate load plants*. They are often run for most of the daytime and then cycled as necessary to follow the evening load.
- ▶ Generation planners are responsible to decide what combination of power plants will most economically meet the hour-by-hour power demands of their customers. This is *unit commitment*.

Unit Commitment

- ▶ Unit Commitment is to commit a generating unit:
 - ▶ bring the unit up to speed;
 - ▶ synchronize it to the system;
 - ▶ connect it to the network.
- ▶ In modern electricity grids, unit commitment has become increasingly complex:
 - ▶ Variable renewable integration: solar PV and wind power's intermittent nature requires more flexible scheduling of conventional generators to compensate for fluctuations.
 - ▶ Computational challenges: it involves thousands of variables and constraints, requiring advanced algorithms like mixed-integer linear programming and heuristic methods.
 - ▶ Multi-objective optimization: beyond cost minimization, modern unit commitment considers emissions reduction, renewable utilization, and grid reliability.
 - ▶ Uncertainty management: stochastic and robust unit commitment approaches account for forecast uncertainties in both demand and renewable generation.

Generation output controllable units

- ▶ Some generators, their outputs can be adjusted within the limits.
- ▶ The status and outputs of these generators can be optimized
 - ▶ Thermal power plants
 - ▶ Coal-fired
 - ▶ Oil-fired
 - ▶ Open cycle gas turbine
 - ▶ Combined cycle gas turbine
 - ▶ Hydropower plant with storage

Generation output uncontrollable units

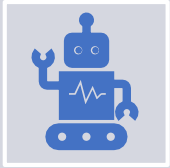
- ▶ Some units, their outputs are hard to control, such as,
 - ▶ Nuclear
 - ▶ Run-of-the-river hydro
 - ▶ Wind turbine
 - ▶ Solar PV
 - ▶ Combined heat and power (CHP, cogeneration)
- ▶ Their outputs are treated as given when doing optimization



Power System Control



Introduction

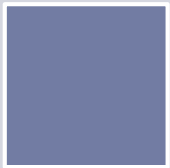


In order to keep the power system in the steady-state, we need



Control of real power

Frequency control (FC), or
Automatic generation
control (AGC)

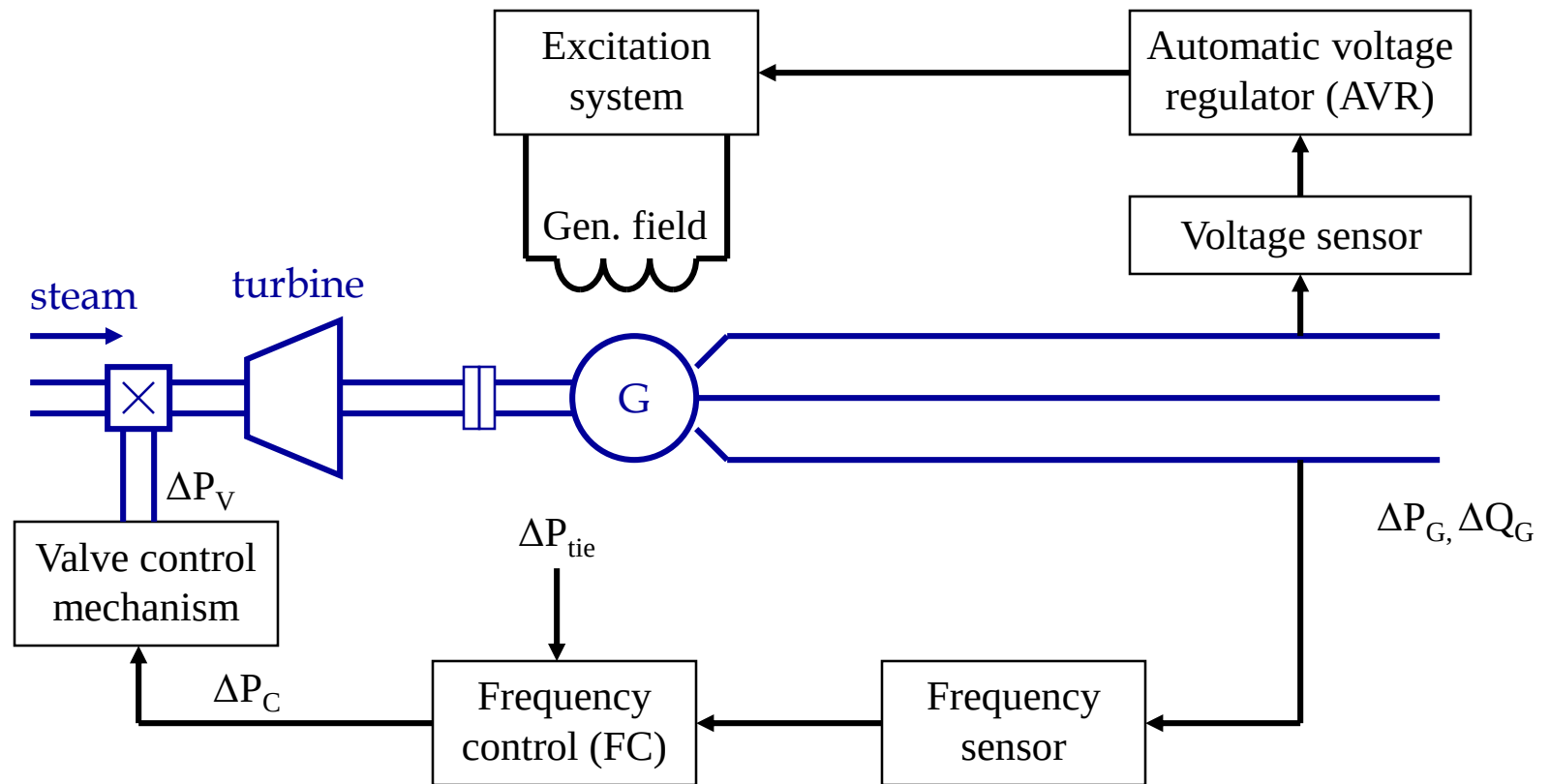


Reactive power control

Voltage control or
Automatic voltage regulator
(AVR)

Basic Generator Control Loops

- Frequency control and voltage control are installed for all generators in an interconnected power system.

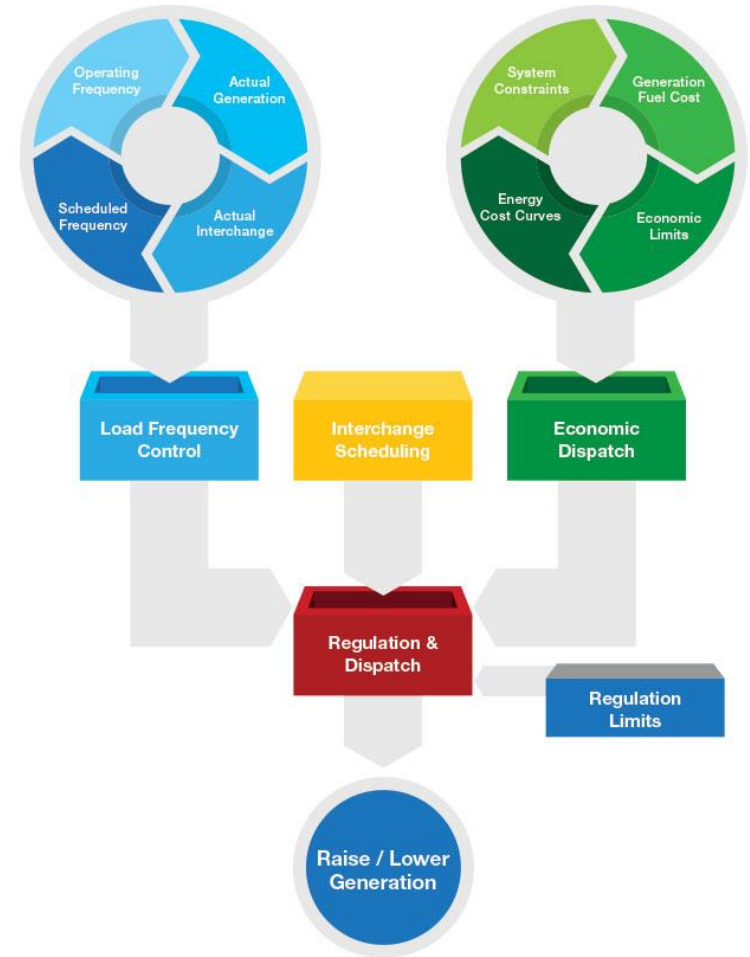


Real Power Control

- ▶ Load of a power system is always changing.
- ▶ Renewable energy generation changes worsen the situation.
- ▶ Generators must respond to the changes to balance the power load.
- ▶ If there is excess generation, generator speeds up, and frequency goes up.
- ▶ Deficiency of generation, speed drops, frequency drops.
- ▶ The deviations from nominal speed or frequency are used as control signals to adjust appropriate valve action automatically.

Functions of Automatic Generation Control (AGC)

- ▶ As the system load changes continuously, the generation is adjusted automatically to restore the frequency to the nominal value.
- ▶ In an interconnected system, the role of AGC is to divide the load among systems and generators so as to achieve maximum economy and tie-line power control.
- ▶ <https://youtu.be/8UreN0bpHjA?si=CSwxl9nQlnqjzWYy>





Operation and Control issues considering Renewable Energy connections



Wind Turbines

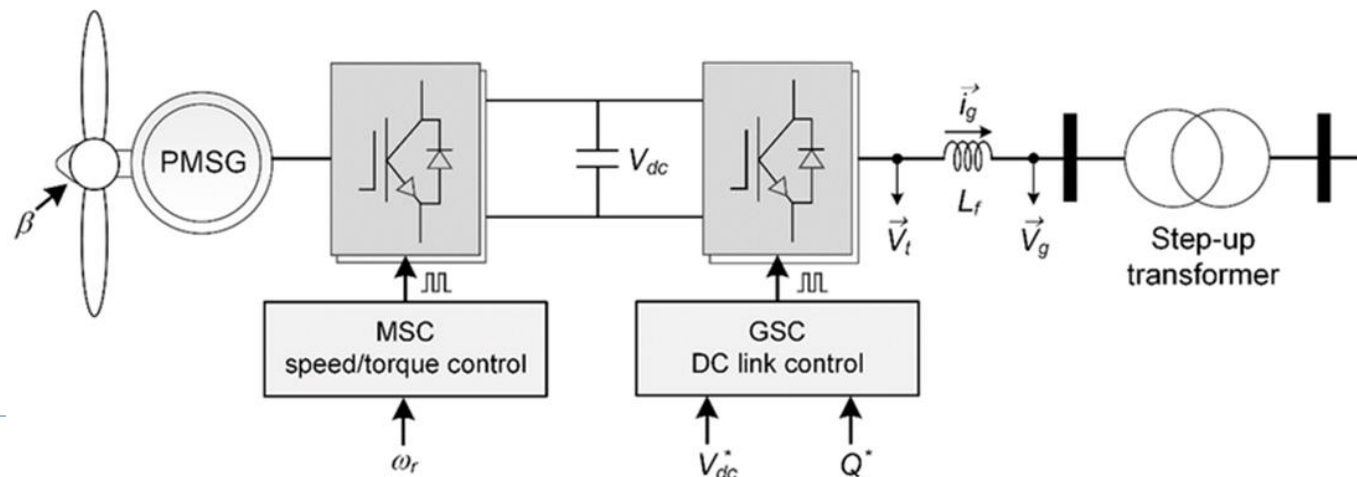
- ▶ There are different types of wind turbines from different manufactures.
- ▶ In these years, there is an obvious trend of using a
 - ▶ Double-feed induction generator, with
 - ▶ Variable speed, and
 - ▶ Variable pitch control.

Types of Wind Turbines

- ▶ To discuss the output fluctuation of wind power, the types of wind turbines need to be discussed.
- ▶ Two major types
 - ▶ Fixed-speed wind turbine
 - ▶ Variable-speed wind turbine

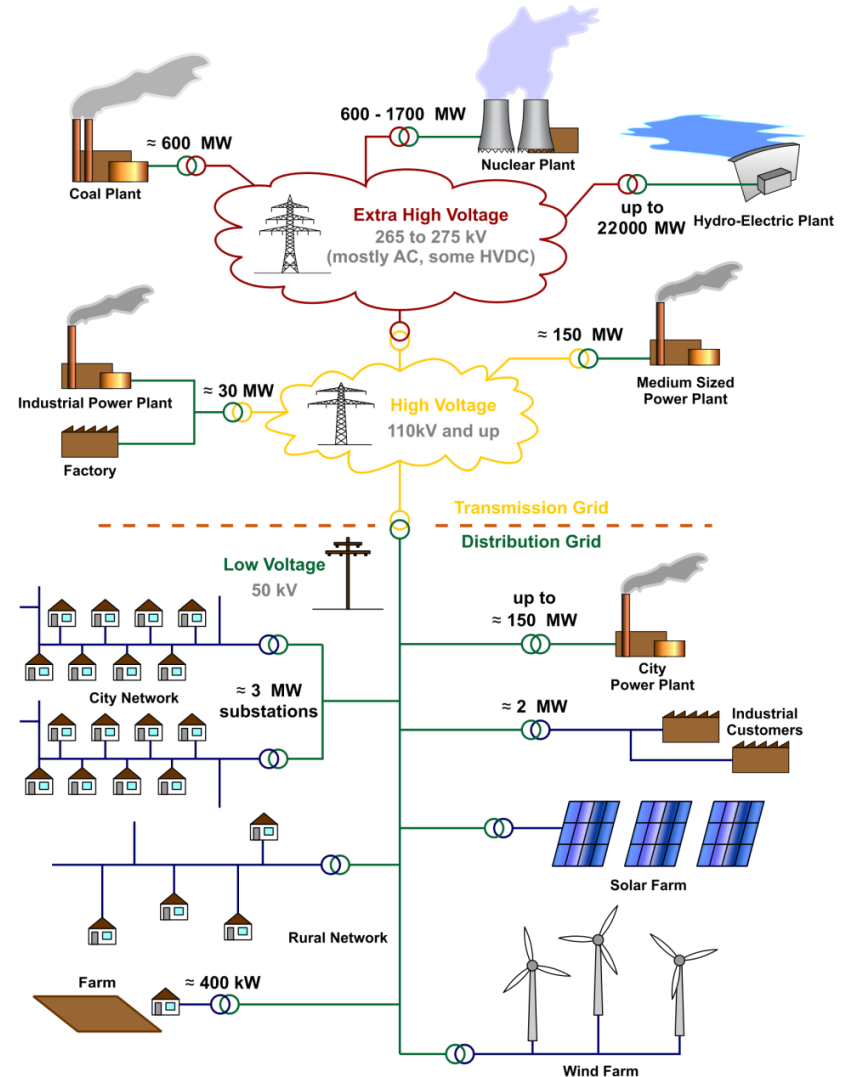
Variable-Speed Wind Turbine

- ▶ Becomes dominant in the past decade.
- ▶ Adapt the rotational speed (ω) of the wind turbine to the wind speed (v).
- ▶ The tip speed ratio is kept constant.
- ▶ A variable-speed system keeps the generator torque fairly constant, and the variations in wind are absorbed by changes in the generator speed.
- ▶ Connected to the grid through power converter.



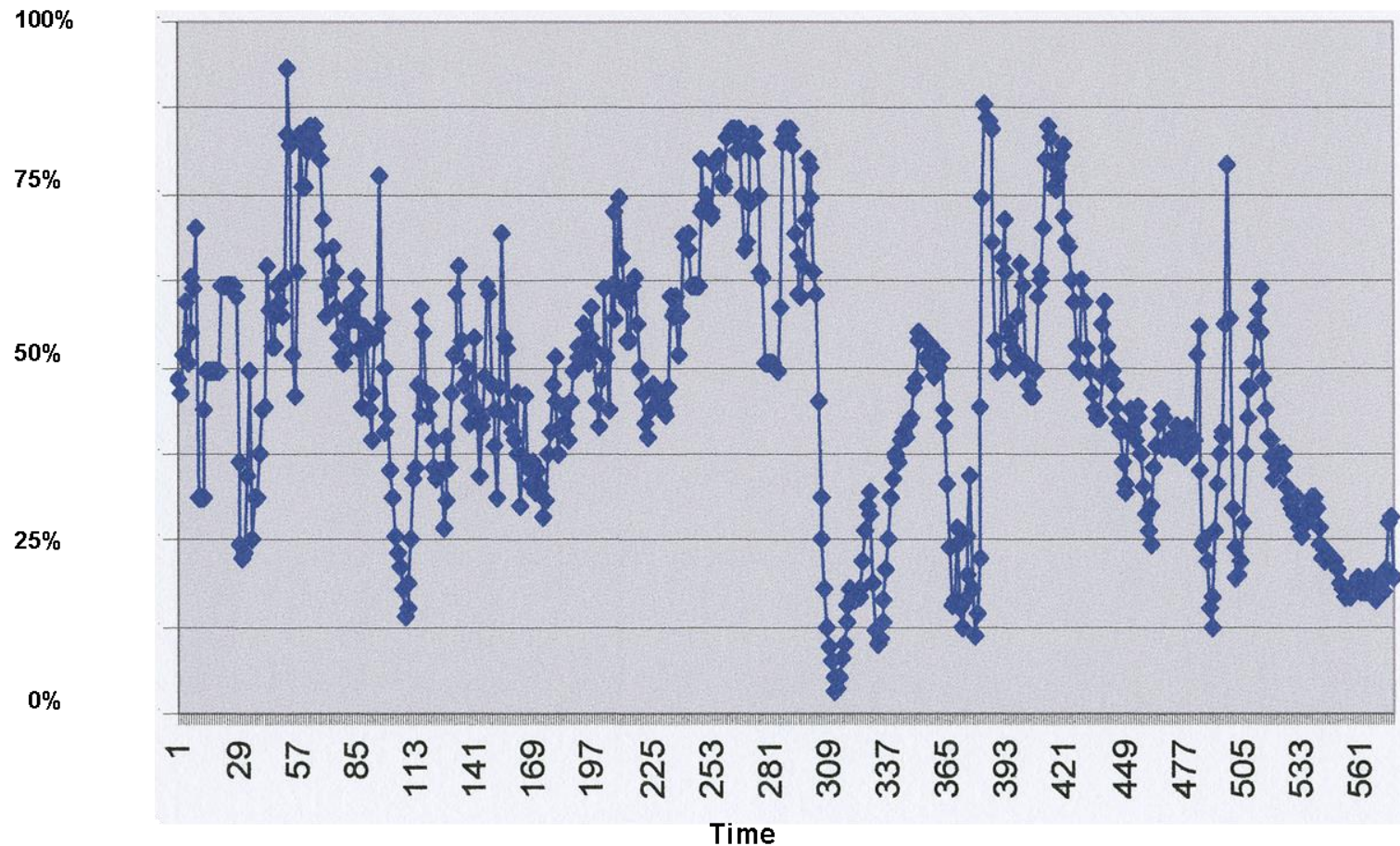
Power Grid

- ▶ In a power grid, the fundamental is the **real-time balance** of power supply and demand.
- ▶ Synchronous AC grid, frequency 50Hz.
- ▶ Frequency control and load following, automatic generation control (AGC).
- ▶ Spinning reserve, backup reserve.



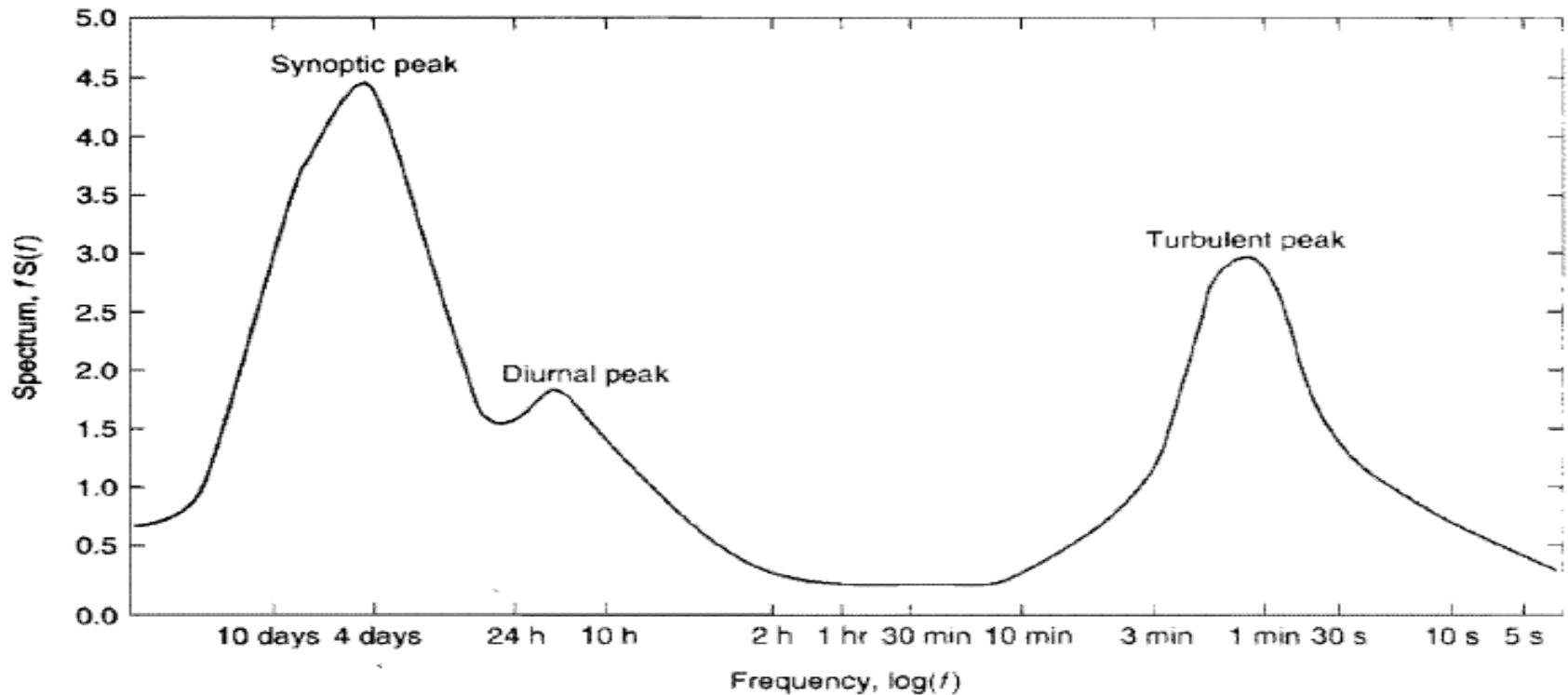
Wind Power Fluctuations

- ▶ A Denmark 160MW offshore wind farm data



5 minute intervals for 48 hours vs. % of wind plant capacity

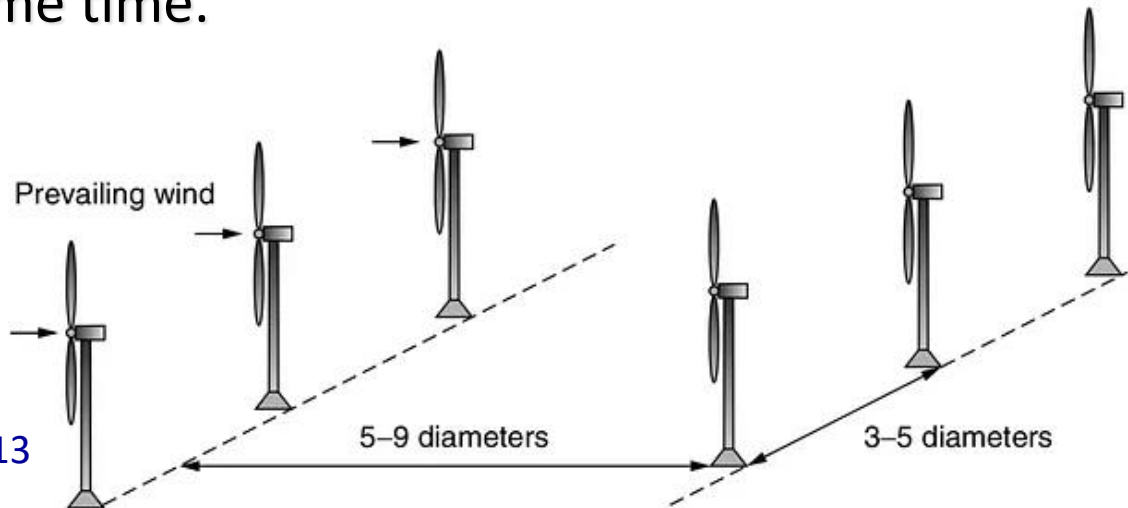
Wind Power Fluctuation – time scale



- ▶ **Turbulent peak** is caused mainly by gusts in the second-minute range
- ▶ **Diurnal peak** depends on daily wind speed variation
- ▶ **Synoptic peak** depends on changing of weather patterns.

Aggregation of Wind Power

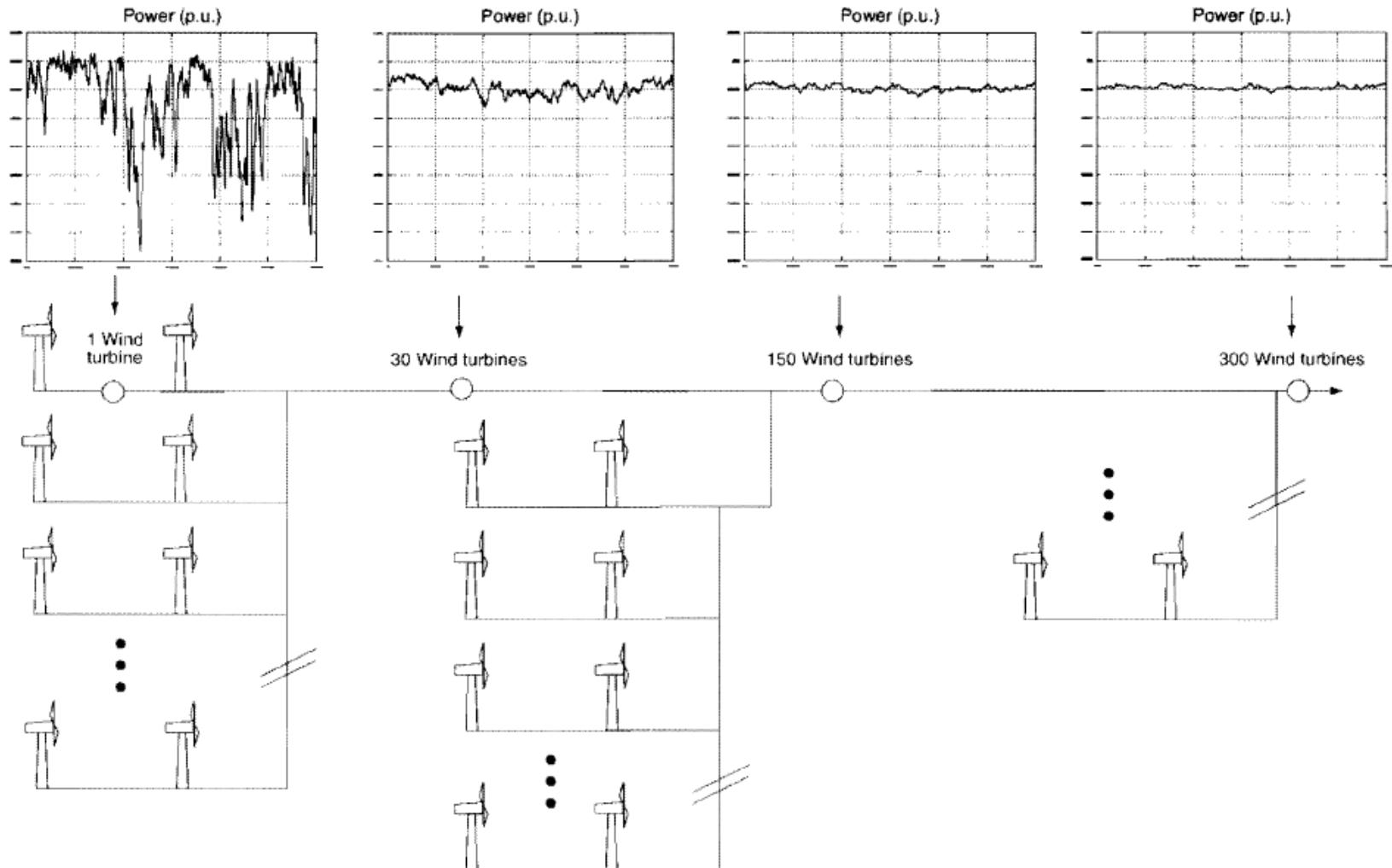
- ▶ Increase the number of wind turbines within a wind farm:
 - ✓ reduce the impact of turbulent peak as gusts do not hit all wind turbines at the same time.



Source: Masters, Gilbert M, 2013

- ▶ Distribute wind farms over a wide geographical area:
 - ✓ reduce the impact of diurnal and synoptic peaks as weather pattern changes do not affect all areas at the same time.

Aggregation of Wind Power



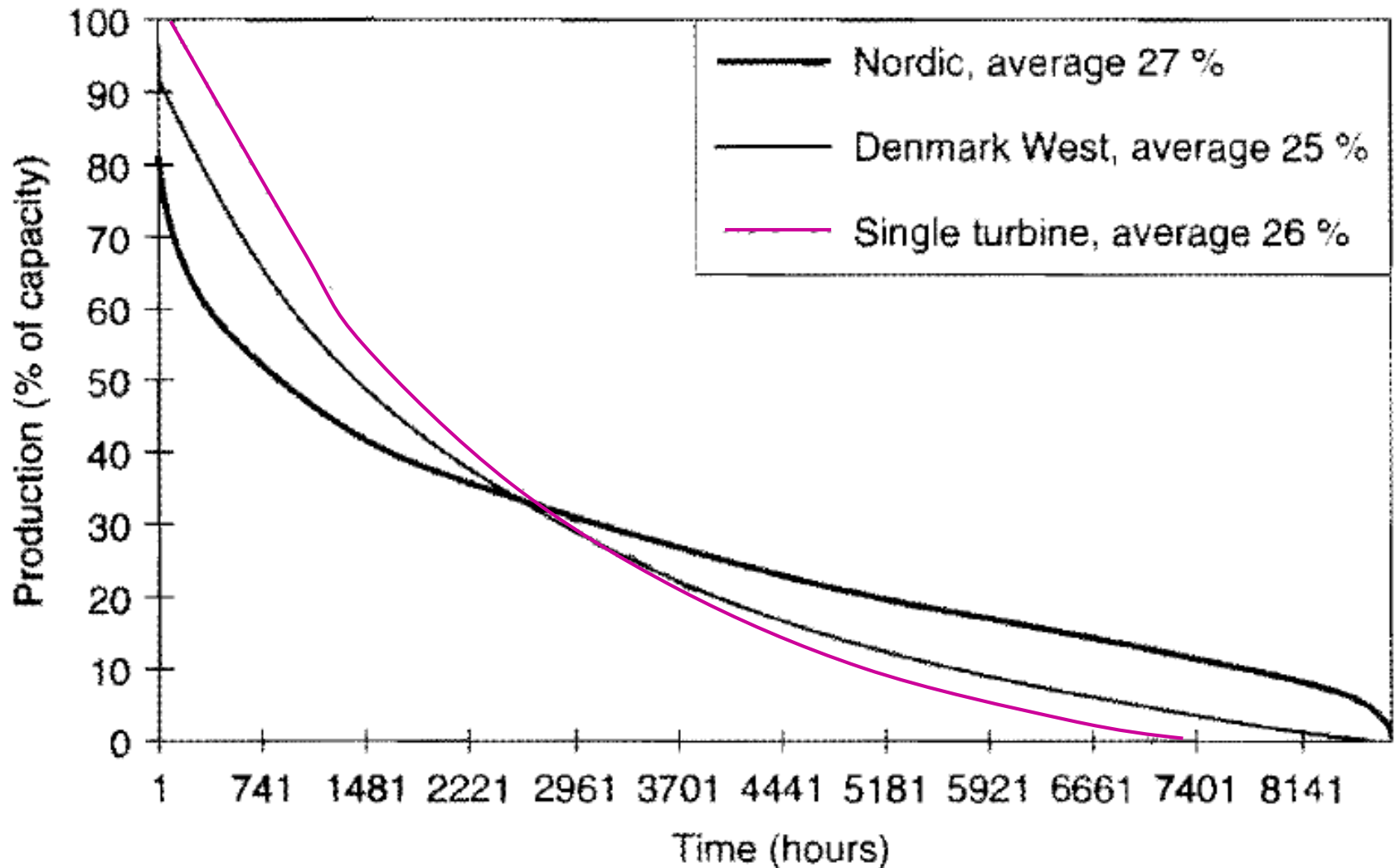
Power System Requirements for Wind Power

Large-scale geographical spreading of wind power will reduce variability.

As wind will not be the same at different sites, for all turbines, there will be fewer instances of near-zero or peak outputs.

For single turbine, output as zero for 10-20% of the time, and nominal output 1-5% of the time.

For large scale wind farm, output is rarely to be zero or nominal value, and it is usually between 5% to 75% of the capacity.



- Duration curves show that increase resources and geographical spreading lead to a flattened duration curve for wind power production.

Source: H. Holttinen, Helsinki Univ. of Tech, 2003

Power variations for a wind farm (extreme situation)

The second-to-second variations will be absorbed in the variable-speed wind turbine.

For a wind farm (103 MW), the extreme ramp rates are

- 4-7% of capacity in one second,
- 10-14% of capacity in one minute, and
- 50-60% of capacity in one hour.

For a large area with geographically dispersed wind farms.

- The second and minute variations will not be significant, and
- Hourly variations will be less than 50~60% of capacity.

Power variation for a large area

- ▶ The largest hourly variations (extreme values) are about
 - ▶ $\pm 30\%$ of capacity for an area around $200 \times 200 \text{ km}^2$ (West or East Denmark)
 - ▶ $\pm 20\%$ of capacity for an area around $400 \times 400 \text{ km}^2$ (Germany, Denmark, Finland or Iowa, USA)
 - ▶ $\pm 10\%$ of capacity in a larger area (for example, Nordic system with four countries) (extreme value), however, most of the time the hourly variations will be within $\pm 10\%$.

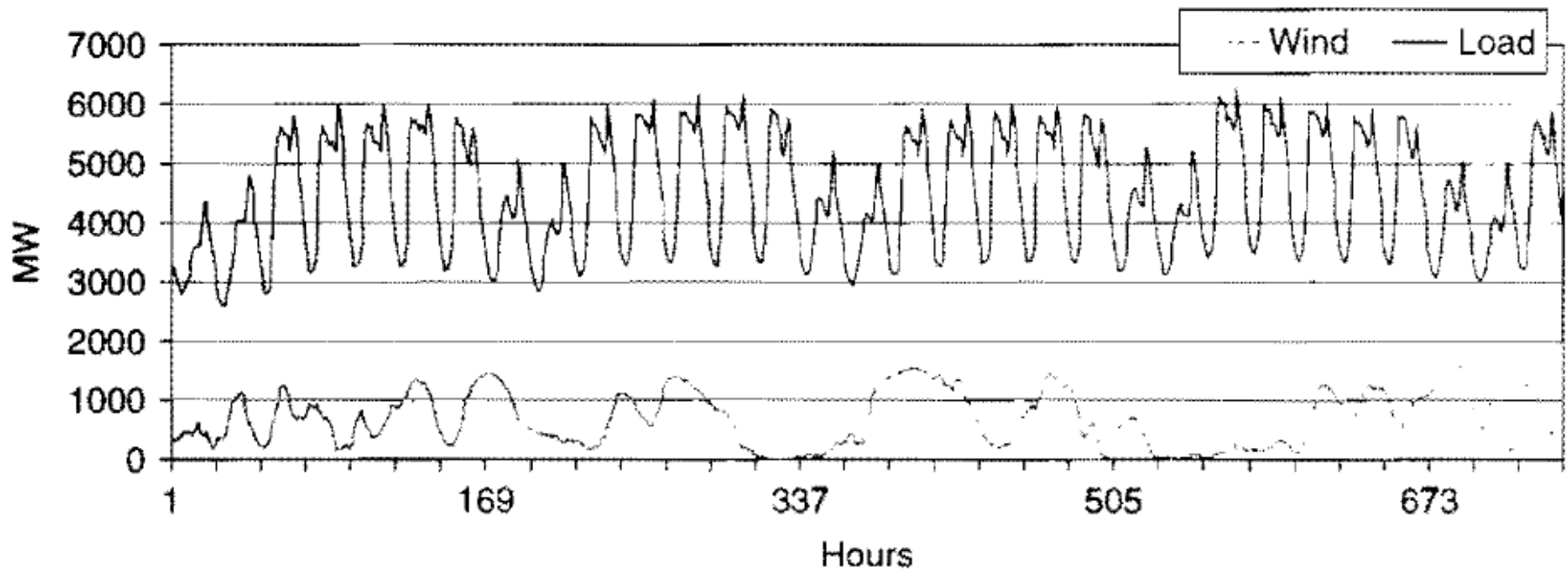
Power System Requirements for Wind Power

Wind resources vary on time scales of seconds, minutes, hours, days, months and years.

Wind production during one year lies between $\pm 15\%$ of the average long-term yearly production.
The yearly variation of hydro power is even larger.

There is a distinct seasonal and daily (diurnal) pattern. They depend on locations, weather and geographical conditions.

Load pattern and wind power output in one month



- ▶ The varying wind resources affect the power system.

Effects of wind energy on the power system

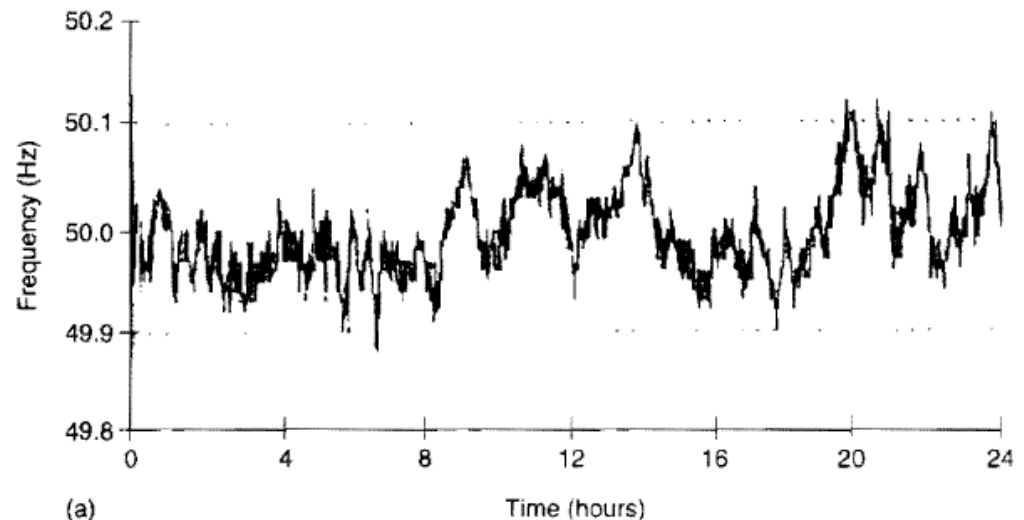
- ▶ Power system operation needs reserves.
 - ▶ Reserve is usually determined according to the largest unit outage.
- ▶ Hourly wind power variation affects the reserves used for frequency control.

Frequency Control

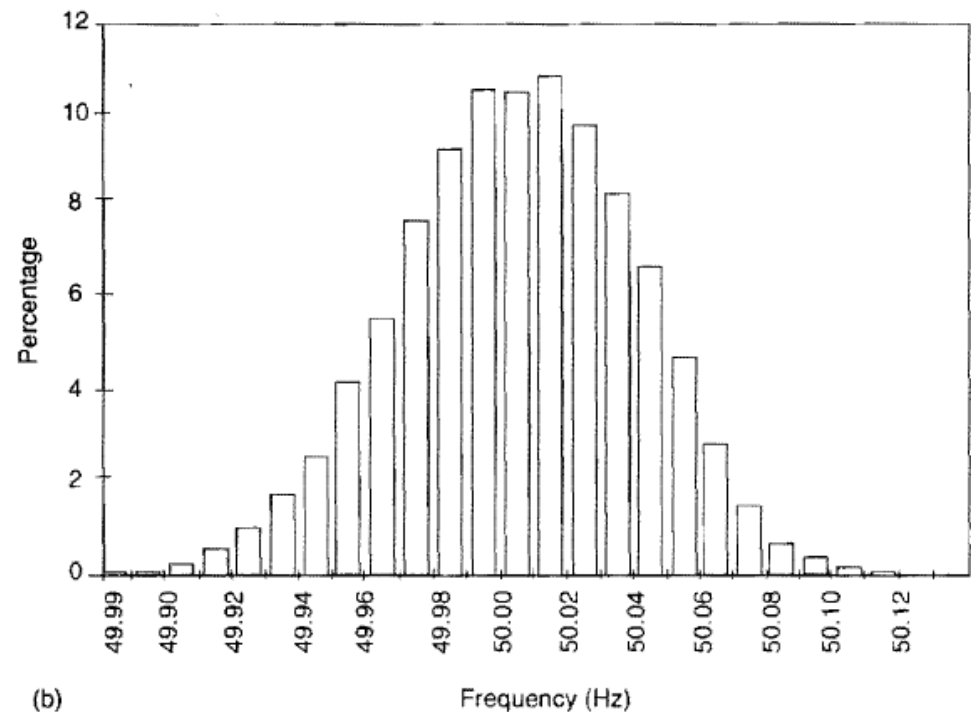
- ▶ In a power system, frequency lies between 50 ± 0.1 Hz, or 50 ± 0.5 Hz
- ▶ Frequency shows the imbalance between supply and demand.
- ▶ Traditional generator has primary control and secondary control to restore frequency and keep the balance of energy after a disturbance.
- ▶ The time span is $1s \sim 10min$

Frequency variations and distributions

(a) Frequency variations in the system during one day;

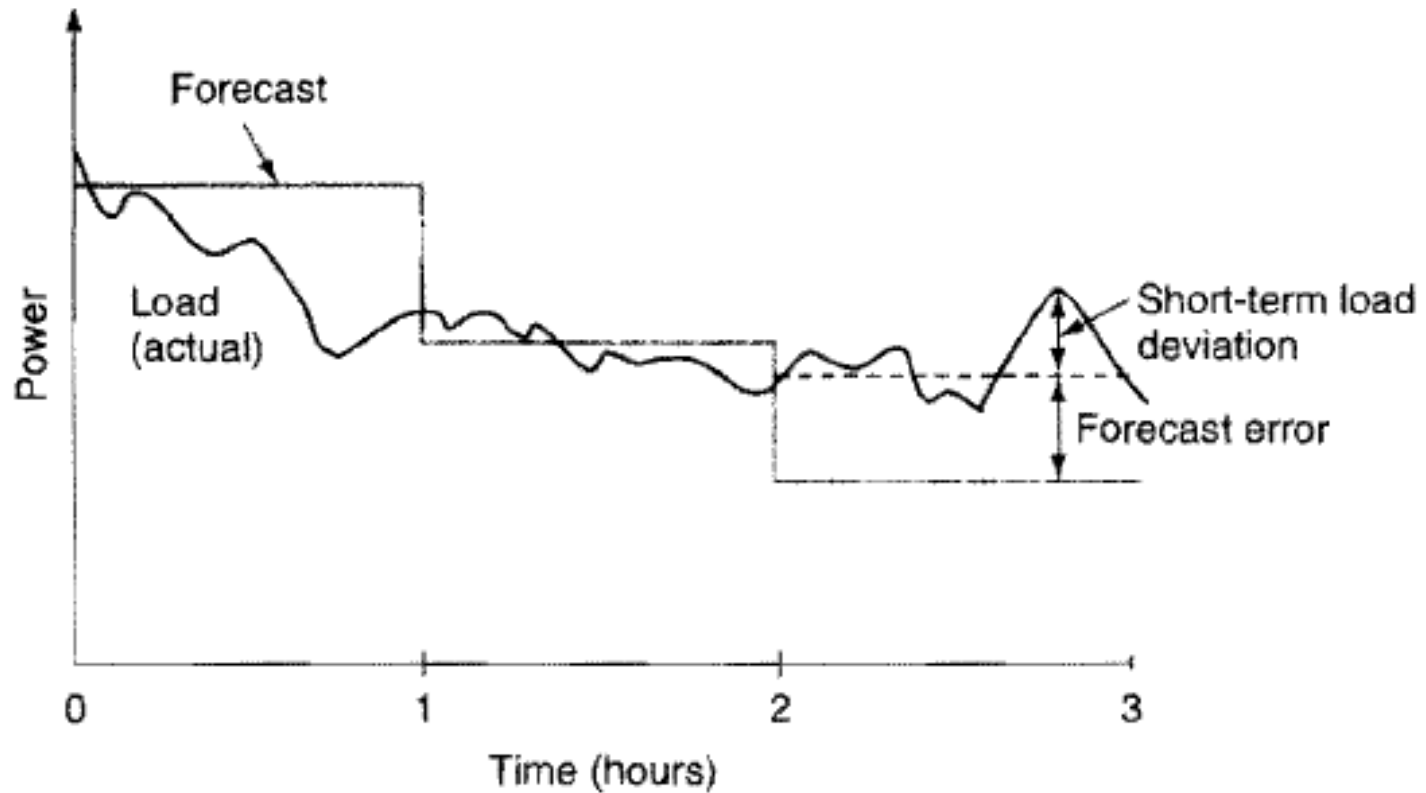


(b) Frequency distribution in the system during one week.



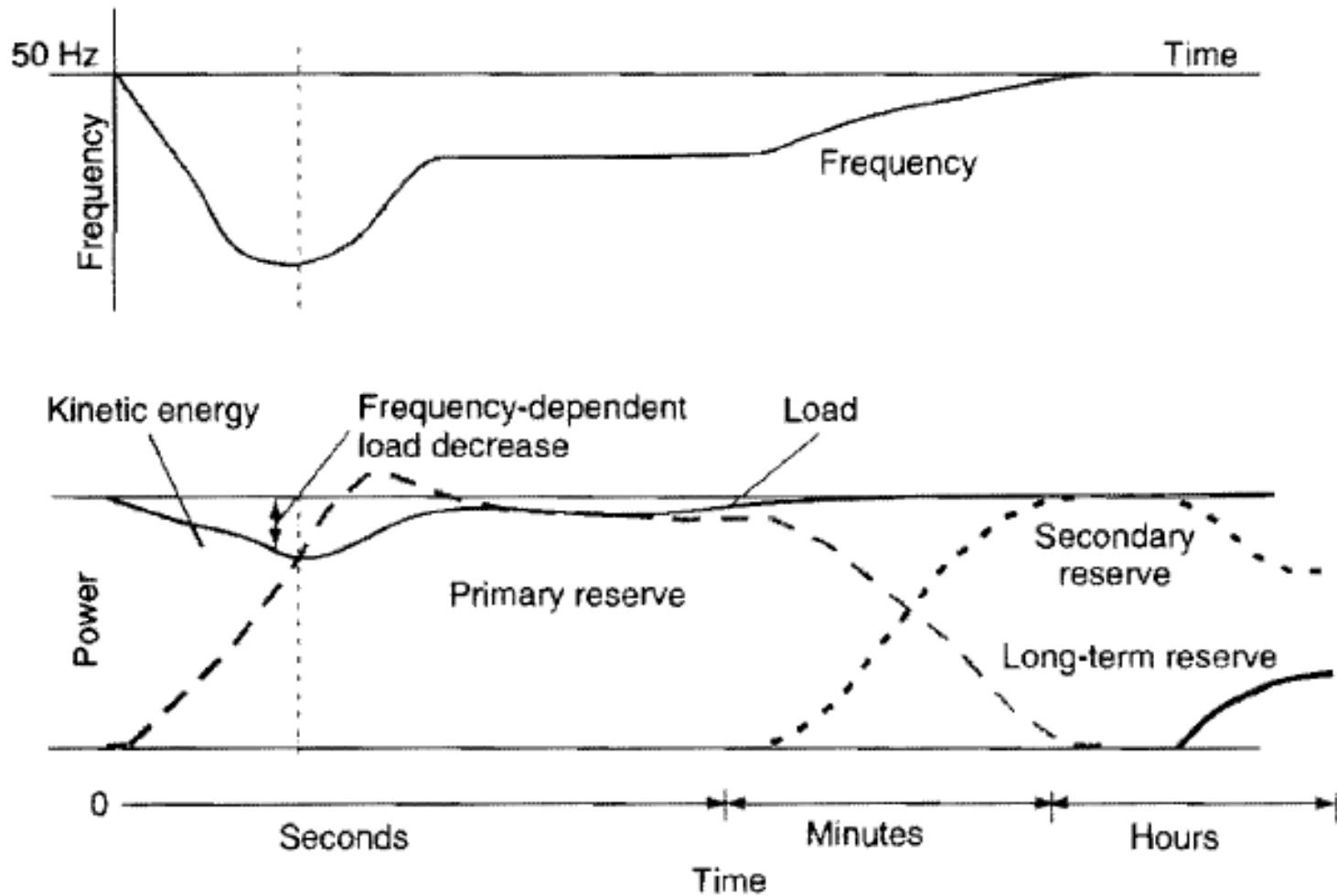
Source: R. Hirvonen, Helsinki Univ. of Tech, 2000

Actual load compare to forecasted load



Source: R. Hirvonen, Hlesiki Univ. of Tech, 2000

Frequency control



Source: R. Hirvonen, Hlesiki Univ. of Tech, 2000

Frequency Control

Power output of a wind farm can vary up to 15% of installed capacity within 15 minutes.

This causes additional imbalance of supply and demand.

However, wind generators use different technologies than conventional generators, they have a limited capability in frequency control.

Wind farms are required to have frequency control capabilities of 3~5% of power output.

Reserve requirement

If wind penetration is high in a system, it requires more reserves to mitigate the variation of wind power generation.

For a 10% penetration level, the extra reserve of 2-8% of the installed wind power capacity is needed (Holttinen, 2003).

Wind Power Penetration

- ▶ Here, wind energy penetration is defined as
The energy produced by wind power (annually) as a percentage of the gross electricity consumption.
- ▶ Low penetration means $< 5\%$ of gross demand
- ▶ High penetration is more than 10%.

Wind Power Penetration

- ▶ Nowadays, wind power penetration is rather limited.
 - ▶ Some systems, Crete in Greece and Denmark is high;
 - ▶ Nordel and Central European countries is low;
 - ▶ The USA is even lower.

Country	Wind Energy Penetration Rate (%)
Denmark	20
Spain	12
Portugal	9
Ireland	8
Germany	7
United States	1

Source: R. Wiser and M. Bolinger, *Annual Report on U.S. Wind Power Installation, Cost, and Performance Trends: 2007*, U.S. DOE, May 2008, p. 6.

Discarded wind energy

- ▶ Due to the intermittent of wind, capacitor factor of wind power generation is much lower than other types of generators.
- ▶ **Capacity Factor** is defined in the next page.
- ▶ For same amount of energy generated, installation capacity of wind power is much higher than conventional power plant.

Capacity Factor of wind turbine / wind farm

- ▶ **Capacity Factor (CF):** P_m / C_{IC}
 - ▶ P_m is power production of a turbine / a farm; C_{IC} is the total capacity of the turbine / wind farm.
- ▶ CF depends on the location and the type of wind turbine. Between 0.25 (low wind speed sites) and 0.4 (high speed sites)
- ▶ Utilization time: $8760 * CF$
 - ▶ Utilization time usually in the range of 2200 – 3500 hours per year.
- ▶ CHP: 4000~5000 h/a, nuclear power 7000~8000 h/a, coal-fired power plant 5000~6000 h/a.

Discarded wind energy

- ▶ If wind power production exceeds the amount that can be safely absorbed by the system, part of the wind energy produced may be discarded -> curtailment.
- ▶ Even with high wind, thermal plants running at partial load to provide frequency regulation.
- ▶ It shows that 10% energy penetration level is the starting point of curtailing wind power.
- ▶ For 20% energy penetration, the discarded energy is about 10% of the total wind power generated.

Operation requirements on Wind Turbines

- ▶ Due to the operation requirement of power systems, power companies set technical requirements for interconnection of wind turbines to power systems
 - ▶ Frequency: Wind farms are required to have frequency control capabilities of 3~5% of power output.
 - ▶ Voltage and tap changers
 - ▶ etc.

Voltage control

- ▶ Power system voltage requirement for each node.
- ▶ Wind farms have to be able to maintain the voltage within $\pm 10\%$ of the reference value.

Tap Changer



It is required that every wind farm that is connected to the network should have an on-load tap-changer.



Tap step requirement

2.5% in a 110kV system

1.6% in 220 ~ 400kV system

Impacts of RE on System Operation

Uncertainties of RE Generations

- Load forecasting as well as RE generation forecasting are needed.
- Requirement of “real-time” control to balance the sudden change of generation.

Power System Operation

- More reserves are needed for a high penetration of intermittent RE generation.
- The reserves should be fast-response turbines or hydro turbine.
- Real-time information exchange and communication infrastructure are needed

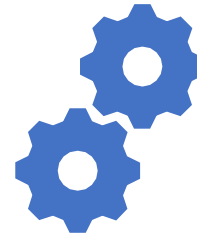
Impacts of RE on System Operation



Economic Dispatch

The operation cost of RE is small, as there is no fuel cost. However, the installation cost is high.

RE is dispatched to generation as much as possible. However, the uncertainties of RE requires a higher reserve backup, which also cost money.



Energy-saving Dispatch

Optimize “energy” / “carbon” consumption.

Optimize “carbon emission”.

The result is, clean energy sources have higher priority to generate; coal-fired power plants with high emissions have low priority, although their operating cost is less expensive.

Impacts of RE on System Control

▶ Frequency control

- ▶ Active power unbalance will cause frequency deviation.
- ▶ Fast-response energy generation devices for frequency control. Including hydro, gas turbine, and energy storage.

▶ Energy storage is a promising technology for frequency control

- ▶ Pump storage, flywheel, battery, etc.

▶ Voltage control

- ▶ Sudden changes of active power outputs from wind farm or solar farm could also cause voltage problem.

Power System Operation and Control



System economic dispatch, frequency and voltage controls are implemented in the *Control Center*.



Control center is the soul of operation of a power grid.



Renovation of control center and its related ICT and control devices is the basic for heading a smart grid.



Generation and demand management, control strategies are the advanced applications.



What is a Smart Grid?



What is a Smart Grid? ([FERC](#))



Self-heals



Motivates and
includes the
consumer



Resists attack



Provides power
quality for 21st
century needs



Accommodates
all generation
and storage
options



Enables markets



Optimizes assets
and operates
efficiently

Incentives for Smart Grid

- ▶ Development of renewable energy generation technology
- ▶ Distributed generation and distributed renewable generation in distribution network (form microgrid)
 - ▶ Bi-directional power flow
- ▶ Demand-side response
 - ▶ Energy saving
 - ▶ Market incentives
- ▶ Long distance power transmission, electricity market
 - ▶ Higher requirement on system monitor and control
 - ▶ Energy exchange, large customer
 - ▶ Wide area energy optimization

Smart Grid Technologies

- ▶ Technologies provide flexibilities
 - ▶ Energy storage, demand response, smart home, virtual power plant, etc.
- ▶ New ICT technologies
 - ▶ Fast and reliable, real-time monitoring, more data, simultaneous, visualization, smarter decision, etc.
- ▶ Power electronic technology, storage tech.
 - ▶ Power flow control
 - ▶ Coordinate generation and storage
- ▶ Upgraded power system automation technology

Smart Grid Today



Smart Meters



Demand side load control



Real-time pricing



Micro-grids



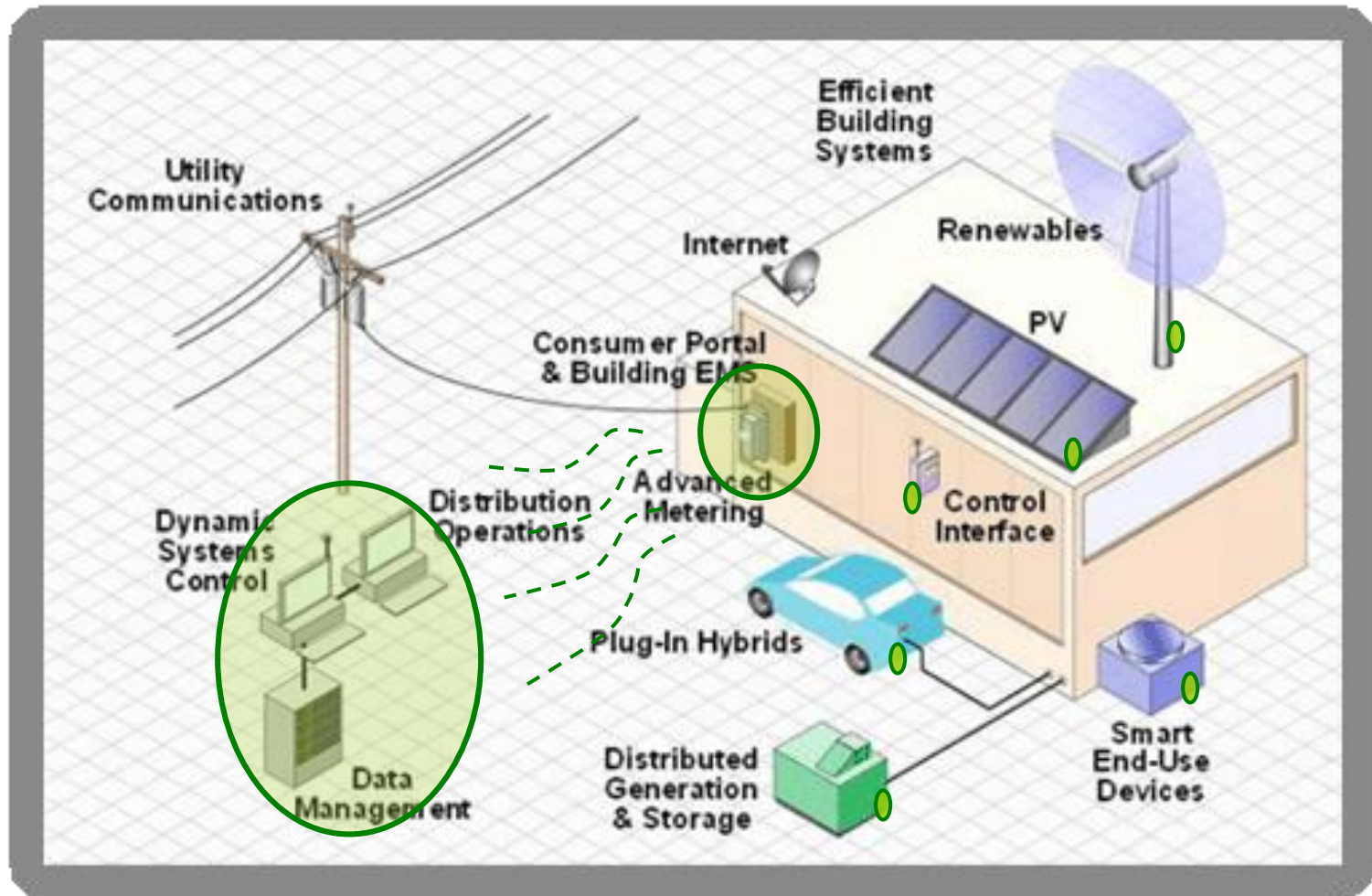
Distribution automation (including two-way flow)

Smart Meter, AMI all over the world



Source: Google map

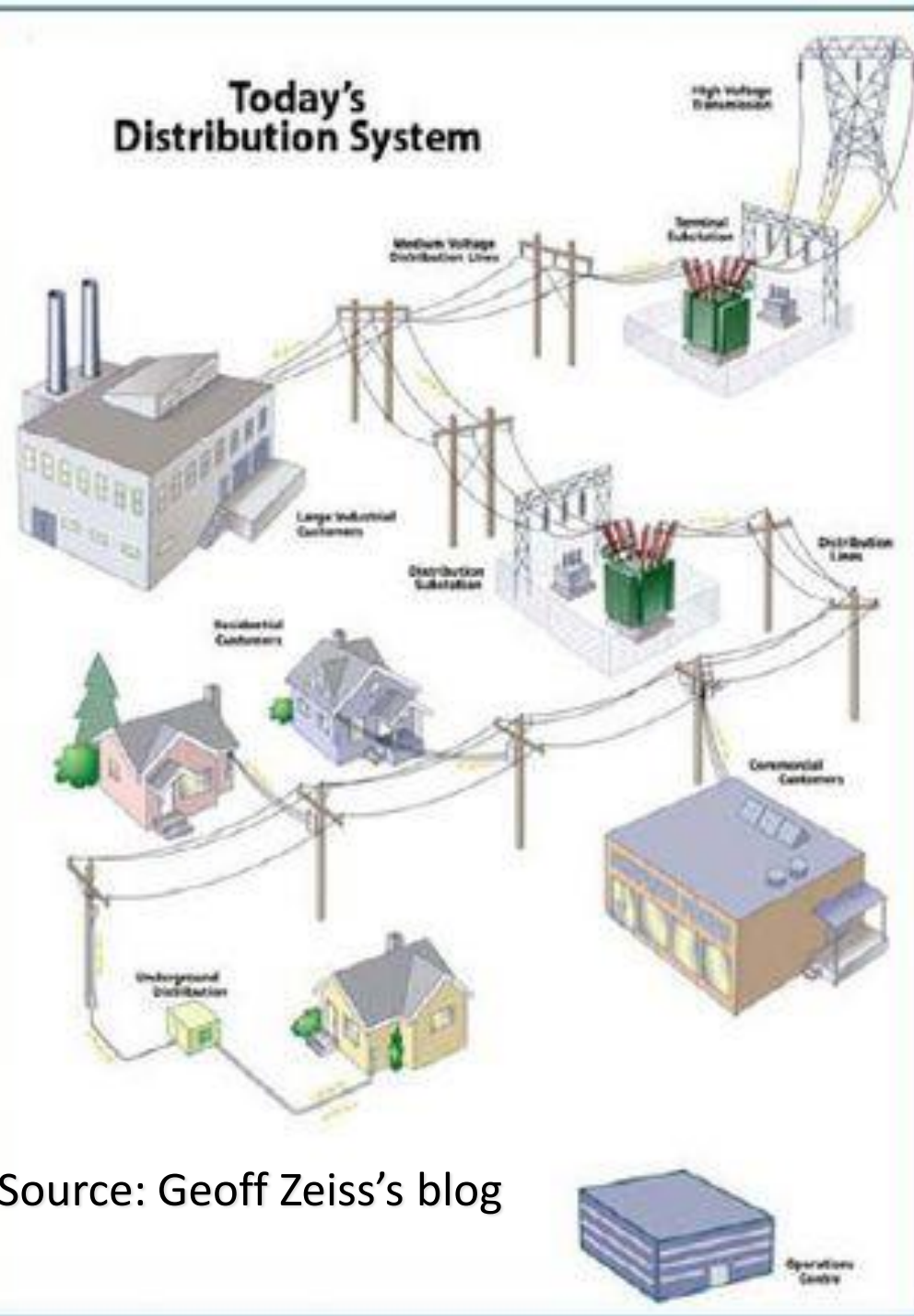
Micro-grid & Demand side management



Distribution Automation

► Distribution Automation (DA) functions implemented are:

- Supervisor control and data acquisition (SCADA)
- Communicating relays in substations
- Remote controlled disconnecting switches
- Distribution management system (DMS) supported by the network information system comprising asset database and mapping system.



Source: Geoff Zeiss's blog

Conclusions

- ▶ Our power grids currently operate on outdated technologies despite advances in automation and communications in other aspects of daily life. While the traditional power grid has served us for over 60 years, several factors are driving necessary change:
 1. transition from fossil fuels to renewable energy sources like solar and wind;
 2. improved battery technology enabling bidirectional electricity flows;
 3. current grids' lack of situational awareness and vulnerability to blackouts;
 4. the urgent need to reduce carbon emissions from power generation.
- ▶ The traditional one-way power flow (from generation plants to customers) is becoming obsolete. The future grid must accommodate bidirectional energy flows between utilities and consumers with distributed energy resources. This evolution requires:
 1. enhanced control and monitoring technologies;
 2. solutions that are reliable, scalable, interoperable, secure and cost-effective;
 3. flexible operation with remote interaction capabilities;
 4. autonomous coordination between grid components like substations.